11.3 /The Integral Test and Estimates of Sums



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The Integral Test and Estimates of Sums

- In general, it is difficult to find the exact sum of a series.
- We were able to accomplish this for geometric series and the series $\Sigma\ 1/[n(n+1)]$

because in each of those cases we could find a simple formula for the nth partial sum s_n.

- But usually it isn't easy to discover such a formula.

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The Integral Test Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x)\,dx$ is convergent. In other words:

(a) If $\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.

(b) If $\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

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THE INTEGRAL TEST AND ESTIMATES OF SUMS

Example: Test the series for convergence or divergence.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \quad \Leftrightarrow \quad \int_{1}^{\infty} \frac{1}{x^2} dx$$

$$\Sigma \approx \int$$

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THE INTEGRAL TEST AND ESTIMATES OF SUMS

THEOREM: (Integral Test) $f(n) = a_n$

$f(x)$ a continuous, positive, decreasing function on [1, inf)

$\sum_{n=1}^{\infty} a_n$ Convergent $\Leftrightarrow$ $\int_1^{\infty} f(x)dx$ Convergent

$\sum_{n=1}^{\infty} a_n$ Divergent $\Leftrightarrow$ $\int_1^{\infty} f(x)dx$ Dinvergent

Example:

Test the series for convergence or divergence.

$$\sum_{n=1}^{\infty} \frac{\ln n}{n}$$

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Example: **Harmonic Series**

is the series convergent?

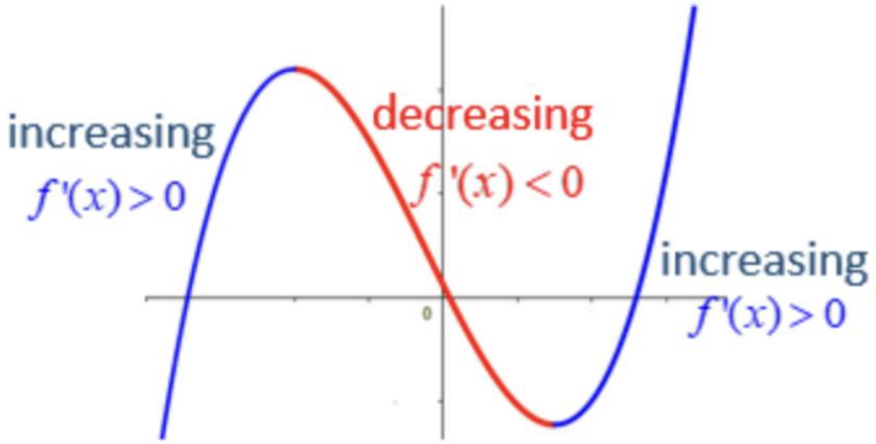
$$\sum_{n=1}^{\infty} \frac{1}{n}$$

Increasing and Decreasing Functions

1.If $f'(x) > 0$ for every x on some interval I, then f(x) is increasing on the interval.

2.If $f'(x) < 0$ for every x on some interval I, then f(x) is decreasing on the interval.

3.If $f'(x) = 0$ for every x on some interval I, then f(x) is constant on the interval.



Memorize:

$$\int_1^{\infty} \frac{1}{x^p}\,dx \quad \text{is convergent if } p > 1 \text{ and divergent if } p \leqslant 1.$$

Example:

For what values of p is the series convergent? $\sum_{n=1}^{\infty} \frac{1}{n^p}$

Special Series:

1) Geometric Series $\sum_{n=1}^{\infty} ar^{n-1}$
2) Harmonic Series $\sum_{n=1}^{\infty} \frac{1}{n}$
3) Telescoping Series $\sum_{n=1}^{\infty} (b_n - b_{n+1})$
4) p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$
5) Alternating p-series

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Facts about: (Harmonic Seris)

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

1)The harmonic series diverges, but very slowly.

the sum of the first million terms is less than **15**

the sum of the first billion terms is less than **22**

2) If we delete from the harmonic series all terms having the digit 9 in the denominator. The resulting series is **convergent.**

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Example 1

- Test the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ for convergence or divergence.

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Example 4

- Determine whether the series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{\ln n}{n}$$

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